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Chapter 3

Rigid-Body Motions

In the previous chapter, we saw that a minimum of six numbers is needed to specify the position and orientation of a rigid body in three-dimensional physical space. In this chapter we develop a systematic way to describe a rigid body's position and orientation which relies on attaching a reference frame to the body. The configuration of this frame with respect to a fixed reference frame is then represented as a 4×4 matrix. This matrix is an example of an implicit representation of the C-space, as discussed in the previous chapter: the actual six-dimensional space of rigid-body configurations is obtained by applying ten constraints to the 16-dimensional space of 4×4 real matrices.

Such a matrix not only represents the configuration of a frame, but can also be used to (1) translate and rotate a vector or a frame, and (2) change the representation of a vector or a frame from coordinates in one frame to coordinates in another frame. These operations can be performed by simple linear algebra, which is a major reason why we choose to represent a configuration as a 4×4 matrix.

The non-Euclidean (i.e., non-"flat") nature of the C-space of positions and orientations leads us to use a matrix representation. A rigid body's velocity, however, can be represented simply as a point in $\mathbb{R}^6$, defined by three angular velocities and three linear velocities, which together we call a **spatial velocity** or **twist**. More generally, even though a robot's C-space may not be a vector space, the set of feasible velocities at any point in the C-space always forms a vector space. For example, consider a robot whose C-space is the sphere S^2: although the C-space is not flat, at any point on the sphere the space of velocities can be thought of as the plane (a vector space) tangent to that point on the sphere.

Any rigid-body configuration can be achieved by starting from the fixed (home) reference frame and integrating a constant twist for a specified time. Such a motion resembles the motion of a screw, rotating about and translating along the same fixed axis. The observation that all configurations can be achieved by a screw motion motivates a six-parameter representation of the configuration called the **exponential coordinates**. The six parameters can be divided into the parameters describing the direction of the screw axis and a scalar to indicate how far the screw motion must be followed to achieve the desired configuration.

This chapter concludes with a discussion of forces. Just as angular and linear velocities are packaged together into a single vector in $\mathbb{R}^6$, moments (torques) and forces are packaged together into a six-vector called a **spatial force** or **wrench**.

To illustrate the concepts and to provide a synopsis of the chapter, we begin with a motivating planar example. Before doing so, we make some remarks about vector notation.

A Word about Vectors and Reference Frames

A **free vector** is a geometric quantity with a length and a direction. Think of it as an arrow in $\mathbb{R}^n$. It is called "free" because it is not necessarily rooted anywhere; only its length and direction matter. A linear velocity can be viewed as a free vector: the length of the arrow is the speed and the direction of the arrow is the direction of the velocity. A free vector is denoted by an upright text symbol, e.g., v.

If a reference frame and length scale have been chosen for the underlying space in which v lies then this free vector can be moved to a position such that the base of the arrow is at the origin without changing the orientation. The free vector v can then be represented by its coordinates in the reference frame. We write the vector in italics, $v \in \mathbb{R}^n$, where v is at the "head" of the arrow in the frame's coordinates. If a different reference frame and length scale are chosen then the representation v will change but the underlying free vector v is unchanged.

In other words, we say that v is **coordinate free**; it refers to a physical quantity in the underlying space, and it does not care how we represent it. However, v is a representation of v that depends on the choice of coordinate frame.

A point p in physical space can also be represented as a vector. Given a choice of reference frame and length scale for physical space, the point p can be represented as a vector from the reference frame origin to p; its vector

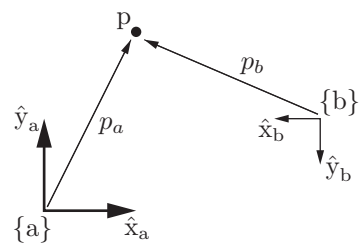


Figure 3.1: The point p exists in physical space, and it does not care how we represent it. If we fix a reference frame {a}, with unit coordinate axes $\hat{x}_a$ and $\hat{y}_a$, we can represent p as $p_a = (1, 2)$. If we fix a reference frame {b} at a different location, a different orientation, and a different length scale, we can represent p as $p_b = (4, -2)$.

representation is denoted in italics by $p \in \mathbb{R}^n$. Here, as before, a different choice of reference frame and length scale for physical space leads to a different representation $p \in \mathbb{R}^n$ for the same point p in physical space. See Figure 3.1.

In the rest of this book, a choice of length scale will always be assumed, but we will be dealing with reference frames at different positions and orientations. A reference frame can be placed anywhere in space, and any reference frame leads to an equally valid representation of the underlying space and the objects in it. We always assume that exactly one stationary **fixed frame**, or **space frame**, denoted {s}, has been defined. This might be attached to a corner of a room, for example. Similarly, we often assume that at least one frame has been attached to some moving rigid body, such as the body of a quadrotor flying in the room. This **body frame**, denoted {b}, is the stationary frame that is coincident with the body-attached frame at any instant.

While it is common to attach the origin of the {b} frame to some important point on the body, such as its center of mass, this is not necessary. The origin of the {b} frame does not even need to be on the physical body itself, as long as its configuration relative to the body, viewed from an observer stationary relative to the body, is constant.

Important! All frames in this book are stationary, inertial, frames. When we refer to a body frame {b}, we mean a motionless frame that is instantaneously coincident with a frame that is fixed to a (possibly moving) body. This is important to keep in mind, since you may have had a dynamics course that used non-inertial moving frames attached to rotating bodies. Do not confuse these with the stationary, inertial, body frames of this book.

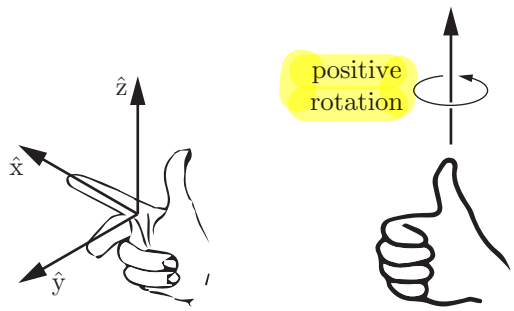


Figure 3.2: (Left) The $\hat{x}$, $\hat{y}$, and $\hat{z}$ axes of a right-handed reference frame are aligned with the index finger, middle finger, and thumb of the right hand, respectively. (Right) A positive rotation about an axis is in the direction in which the fingers of the right hand curl when the thumb is pointed along the axis.

For simplicity, we will usually refer to a body frame as a frame attached to a moving rigid body. Despite this, at any instant, by "body frame" we actually mean the stationary frame that is instantaneously coincident with the frame moving along with the body.

It is worth repeating one more time: **all frames are stationary**.

All reference frames are **right-handed**, as illustrated in Figure 3.2. A positive rotation about an axis is defined as the direction in which the fingers of the right hand curl when the thumb is pointed along the axis (Figure 3.2).

3.1 Rigid-Body Motions in the Plane

Consider the planar body (the gray shape) in Figure 3.3; its motion is confined to the plane. Suppose that a length scale and a fixed reference frame $\{s\}$ have been chosen as shown, with unit axes $\hat{x}_s$ and $\hat{y}_s$. (Throughout this book, the hat notation indicates a unit vector.) Similarly, we attach a reference frame with unit axes $\hat{x}_b$ and $\hat{y}_b$ to the planar body. Because this frame moves with the body, it is called the body frame and is denoted $\{b\}$.

To describe the configuration of the planar body, only the position and orientation of the body frame with respect to the fixed frame need to be specified. The body-frame origin p can be expressed in terms of the coordinate axes of

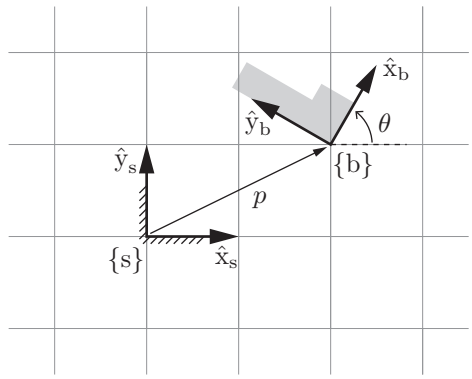


Figure 3.3: The body frame $\{b\}$ is expressed in the fixed-frame coordinates $\{s\}$ by the vector p and the directions of the unit axes $\hat{x}_b$ and $\hat{y}_b$. In this example, $p = (2, 1)$ and $\theta = 60°$, so $\hat{x}_b = (\cos\theta, \sin\theta) = (0.5, 1/\sqrt{2})$ and $\hat{y}_b = (-\sin\theta, \cos\theta) = (-1/\sqrt{2}, 0.5)$.

$\{s\}$ as

$$p = p_x \hat{x}_s + p_y \hat{y}_s. \tag{3.1}$$

You are probably more accustomed to writing this vector as simply $p = (p_x, p_y)$; this is fine when there is no possibility of ambiguity about reference frames, but writing p as in Equation (3.1) clearly indicates the reference frame with respect to which (p_x, p_y) is defined.

The simplest way to describe the orientation of the body frame $\{b\}$ relative to the fixed frame $\{s\}$ is by specifying the angle θ, as shown in Figure 3.3. Another (admittedly less simple) way is to specify the directions of the unit axes $\hat{x}_b$ and $\hat{y}_b$ of $\{b\}$ relative to $\{s\}$, in the form

$$\hat{x}_b = \cos\theta\, \hat{x}_s + \sin\theta\, \hat{y}_s, \tag{3.2}$$

$$\hat{y}_b = -\sin\theta\, \hat{x}_s + \cos\theta\, \hat{y}_s. \tag{3.3}$$

At first sight this seems to be a rather inefficient way of representing the body-frame orientation. However, imagine if the body were to move arbitrarily in three-dimensional space; a single angle θ would not suffice to describe the orientation of the displaced reference frame. We would actually need three angles, but it is not yet clear how to define an appropriate set of three angles. However, expressing the directions of the coordinate axes of $\{b\}$ in terms of coefficients of the coordinate axes of $\{s\}$, as we have done above for the planar case, is straightforward.

Assuming we agree to express everything in terms of $\{s\}$ then, just as the

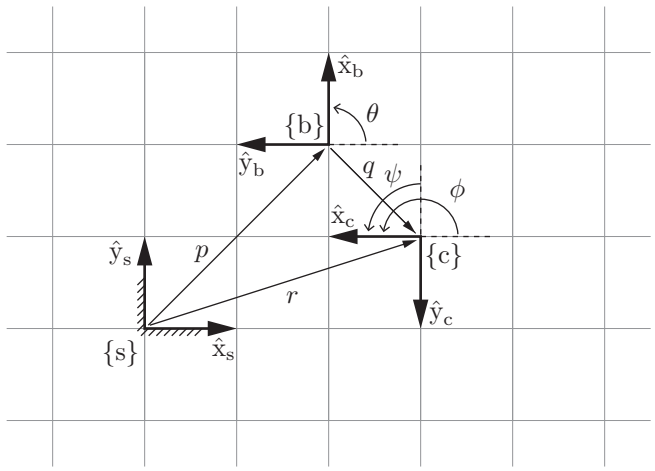


Figure 3.4: The frame {b} in {s} is given by (P, p), and the frame {c} in {b} is given by (Q, q). From these we can derive the frame {c} in {s}, described by (R, r). The numerical values of the vectors p, q, and r and the coordinate-axis directions of the three frames are evident from the grid of unit squares.

point p can be represented as a column vector $p \in \mathbb{R}^2$ of the form

$$p = \begin{bmatrix} p_x \\ p_y \end{bmatrix}, \tag{3.4}$$

位置向量 → 2 dof

the two vectors $\hat{x}_b$ and $\hat{y}_b$ can also be written as column vectors and packaged into the following 2×2 matrix P:

$$P = [\hat{x}_b \;\; \hat{y}_b] = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}. \tag{3.5}$$

旋转矩阵

4个变量 - 3个约束 = 1 dof (θ)

The matrix P is an example of a **rotation matrix**. Although P consists of four numbers, they are subject to three constraints (each column of P must be a unit vector, and the two columns must be orthogonal to each other), and the one remaining degree of freedom is parametrized by θ. Together, the pair (P, p) provides a description of the orientation and position of {b} relative to {s}.

3个约束
P的列必是单位向量
且两列互相垂直

Now refer to the three frames in Figure 3.4. Repeating the approach above, and expressing {c} in {s} as the pair (R, r), we can write

$$r = \begin{bmatrix} r_x \\ r_y \end{bmatrix}, \qquad R = \begin{bmatrix} \cos\phi & -\sin\phi \\ \sin\phi & \cos\phi \end{bmatrix}. \tag{3.6}$$

We could also describe the frame {c} relative to {b}. Letting q denote the vector from the origin of {b} to the origin of {c} expressed in {b} coordinates, and letting Q denote the orientation of {c} relative to {b}, we can write {c} relative to {b} as the pair (Q, q), where

$$q = \begin{bmatrix} q_x \\ q_y \end{bmatrix}, \qquad Q = \begin{bmatrix} \cos\psi & -\sin\psi \\ \sin\psi & \cos\psi \end{bmatrix}. \tag{3.7}$$

If we know (Q, q) (the configuration of {c} relative to {b}) and (P, p) (the configuration of {b} relative to {s}), we can compute the configuration of {c} relative to {s} as follows:

$$R = PQ \quad \text{(convert } Q \text{ to the \{s\} frame)} \tag{3.8}$$

$$r = Pq + p \quad \text{(convert } q \text{ to the \{s\} frame and vector-sum with } p\text{).} \tag{3.9}$$

Thus (P, p) not only represents a configuration of {b} in {s}; it can also be used to convert the representation of a point or frame from {b} coordinates to {s} coordinates.

坐标变换的链式法则

Now consider a rigid body with two frames attached to it, {d} and {c}. The frame {d} is initially coincident with {s}, and {c} is initially described by (R, r) in {s} (Figure 3.5(a)). Then the body is moved in such a way that {d} moves to {d′}, becoming coincident with a frame {b} described by (P, p) in {s}. Where does {c} end up after this motion? Denoting the configuration of the new frame {c′} as (R', r'), you can verify that

$$R' = PR, \tag{3.10}$$

$$r' = Pr + p, \tag{3.11}$$

which is similar to Equations (3.8) and (3.9). The difference is that (P, p) is expressed in the same frame as (R, r), so the equations are not viewed as a change of coordinates, but instead as a **rigid-body displacement** (also known as a **rigid-body motion**): in Figure 3.5(a) transformation ① rotates {c} according to P and transformation ② translates it by p in {s}.

刚体运动公式

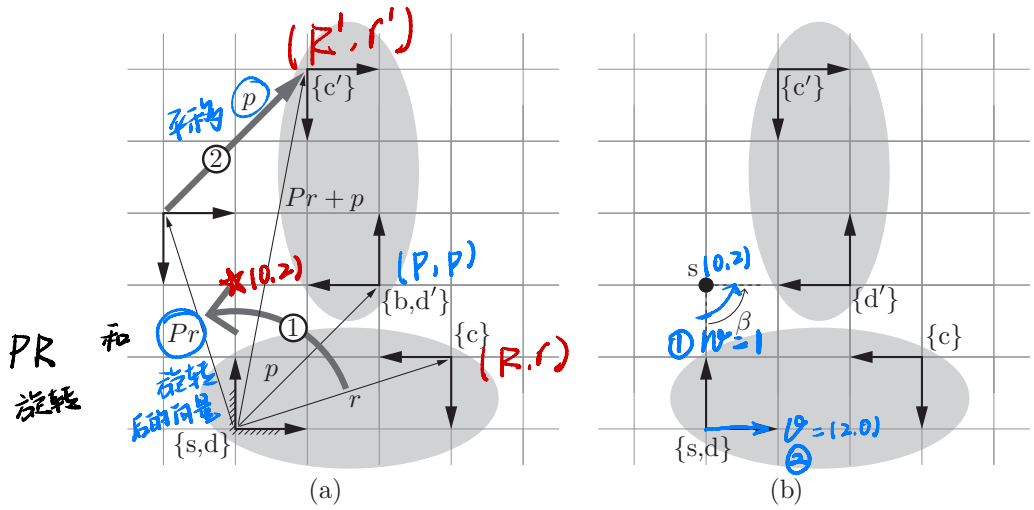
刚体位移

Thus we see that a rotation matrix–vector pair such as (P, p) can be used for three purposes:

(a) to represent a configuration of a rigid body in {s} (Figure 3.3);

(b) to change the reference frame in which a vector or frame is represented (Figure 3.4);

(c) to displace a vector or a frame (Figure 3.5(a)).



PR 旋转 和

Figure 3.5: (a) The frame {d}, fixed to an elliptical rigid body and initially coincident with {s}, is displaced to {d′} (which is coincident with the stationary frame {b}), by first rotating according to P then translating according to p, where (P, p) is the representation of {b} in {s}. The same transformation takes the frame {c}, also attached to the rigid body, to {c′}. The transformation marked ① rigidly rotates {c} about the origin of {s}, and then transformation ② translates the frame by p expressed in {s}. (b) Instead of viewing this displacement as a rotation followed by a translation, both rotation and translation can be performed simultaneously. The displacement can be viewed as a rotation of $\beta = 90°$ about a fixed point s.

刚体运动
先旋转，再平移
=
用刚体绕固定点s转角度β
=
一边转，一边平移
(指数坐标)
螺旋运动
screw motion

Referring to Figure 3.5(b), note that the rigid-body motion illustrated in Figure 3.5(a), expressed as a rotation followed by a translation, can be obtained by simply rotating the body about a fixed point s by an angle β. This is a planar example of a **screw motion**.[1] The displacement can therefore be parametrized by the three screw coordinates (β, s_x, s_y), where $(s_x, s_y) = (0, 2)$ denotes the coordinates for the point s (i.e., the screw axis out of the page) in the fixed frame {s}.

Another way to represent the screw motion is to consider it as the displacement obtained by following simultaneous angular and linear velocities for a given distance. Inspecting Figure 3.5(b), we see that rotating about s with a unit angular velocity ($\omega = 1$ rad/s) means that a point at the origin of the {s} frame moves at two units per second initially in the $+\hat{x}$-direction of the {s} frame, i.e., $v = (v_x, v_y) = (2, 0)$. We can package these together in the three-

旋量引入
(twist)

[1] If the displacement is a pure translation without rotation, then s lies at infinity.

单位螺旋轴 S

第一种：点 s + 角度 β

vector $\mathcal{S} = (\omega, v_x, v_y) = (1, 2, 0)$, a representation of the **screw axis**. Following this screw axis for an angle $\theta = \pi/2$ yields the final displacement. Thus we can represent the displacement using the three coordinates $\mathcal{S}\theta = (\pi/2, \pi, 0)$. These coordinates have some advantages, and we call these the **exponential coordinates** for the planar rigid-body displacement.

第二种：$S = (\omega, v_x, v_y)$

角速度　原点的线速度 初始

S 单位速度

☆ $\dot{\theta}$ 是标量

速度　位移

To represent the combination of an angular and a linear velocity, called a **twist**, we take a screw axis $\mathcal{S} = (\omega, v_x, v_y)$, where $\omega = 1$, and scale it by multiplying by some rotation speed, $\dot{\theta}$. The twist is $\mathcal{V} = \mathcal{S}\dot{\theta}$. The net displacement obtained by rotating about the screw axis $\mathcal{S}$ by an angle θ is equivalent to the displacement obtained by rotating about $\mathcal{S}$ at a speed $\dot{\theta} = \theta$ for unit time, so $\mathcal{V} = \mathcal{S}\dot{\theta}$ can also be considered a set of exponential coordinates.

☆ 不需要式知道 S 坐标

$S\theta$ 指数坐标

$S\dot{\theta}$ 旋量 twist

Preview of the remainder of this chapter. In the rest of this chapter we generalize the concepts above to three-dimensional rigid-body motions. For this purpose consider a rigid body occupying three-dimensional physical space, as shown in Figure 3.6. Assume that a length scale for physical space has been chosen, and that both the fixed frame {s} and body frame {b} have been chosen as shown. Throughout this book all reference frames are right-handed – the unit axes $\{\hat{\mathrm{x}}, \hat{\mathrm{y}}, \hat{\mathrm{z}}\}$ always satisfy $\hat{\mathrm{x}} \times \hat{\mathrm{y}} = \hat{\mathrm{z}}$. Denote the unit axes of the fixed frame by $\{\hat{\mathrm{x}}_\mathrm{s}, \hat{\mathrm{y}}_\mathrm{s}, \hat{\mathrm{z}}_\mathrm{s}\}$ and the unit axes of the body frame by $\{\hat{\mathrm{x}}_\mathrm{b}, \hat{\mathrm{y}}_\mathrm{b}, \hat{\mathrm{z}}_\mathrm{b}\}$. Let p denote the vector from the fixed-frame origin to the body-frame origin. In terms of the fixed-frame coordinates, p can be expressed as

$$p = p_1\hat{\mathrm{x}}_\mathrm{s} + p_2\hat{\mathrm{y}}_\mathrm{s} + p_3\hat{\mathrm{z}}_\mathrm{s}. \tag{3.12}$$

The axes of the body frame can also be expressed as

$$\begin{aligned} \hat{\mathrm{x}}_\mathrm{b} &= r_{11}\hat{\mathrm{x}}_\mathrm{s} + r_{21}\hat{\mathrm{y}}_\mathrm{s} + r_{31}\hat{\mathrm{z}}_\mathrm{s}, && (3.13)\\ \hat{\mathrm{y}}_\mathrm{b} &= r_{12}\hat{\mathrm{x}}_\mathrm{s} + r_{22}\hat{\mathrm{y}}_\mathrm{s} + r_{32}\hat{\mathrm{z}}_\mathrm{s}, && (3.14)\\ \hat{\mathrm{z}}_\mathrm{b} &= r_{13}\hat{\mathrm{x}}_\mathrm{s} + r_{23}\hat{\mathrm{y}}_\mathrm{s} + r_{33}\hat{\mathrm{z}}_\mathrm{s}. && (3.15) \end{aligned}$$

Defining $p \in \mathbb{R}^3$ and $R \in \mathbb{R}^{3\times 3}$ as

$$p = \begin{bmatrix} p_1 \\ p_2 \\ p_3 \end{bmatrix}, \qquad R = [\hat{\mathrm{x}}_\mathrm{b} \;\; \hat{\mathrm{y}}_\mathrm{b} \;\; \hat{\mathrm{z}}_\mathrm{b}] = \begin{array}{c} \begin{array}{ccc} \hat{x}_b & \hat{y}_b & \hat{z}_b \end{array} \\ \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix} \end{array}, \tag{3.16}$$

the 12 parameters given by (R, p) then provide a description of the position and orientation of the rigid body relative to the fixed frame.

Since the orientation of a rigid body has three degrees of freedom, only three of the nine entries in R can be chosen independently. One three-parameter

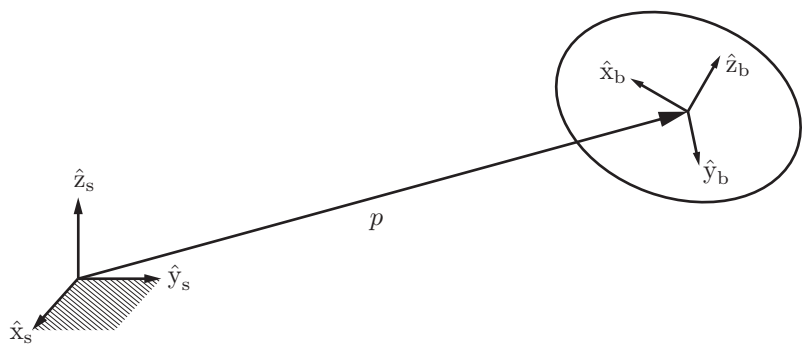


Figure 3.6: Mathematical description of position and orientation.

representation of rotations is provided by the exponential coordinates, which define an axis of rotation and the angle rotated about that axis. We leave other popular representations of orientations (the three-parameter **Euler angles** and the **roll–pitch–yaw angles**, the **Cayley–Rodrigues parameters**, and the **unit quaternions**, which use four variables subject to one constraint) to Appendix B.

We then examine the six-parameter exponential coordinates for the configuration of a rigid body that arise from integrating a six-dimensional twist consisting of the body's angular and linear velocities. This representation follows from the Chasles–Mozzi theorem which states that every rigid-body displacement can be obtained by a finite rotation and translation about a fixed screw axis.

We conclude with a discussion of forces and moments. Rather than treat these as separate three-dimensional quantities, we merge the moment and force vectors into a six-dimensional **wrench**. The twist and wrench, and rules for manipulating them, form the basis for the kinematic and dynamic analyses in subsequent chapters.

3.2 Rotations and Angular Velocities

3.2.1 Rotation Matrices

We argued earlier that, of the nine entries in the rotation matrix R, only three can be chosen independently. We begin by expressing a set of six explicit constraints on the entries of R. Recall that the three columns of R correspond to